Change the following suffix formula to prefix form:

$P \neg Q \wedge \neg RS \vee T \wedge \forall$

5

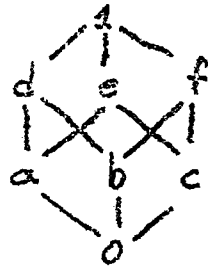
2. Show that the set of Real numbers (between 0 and 1) is not equivalent to the set of Natural numbers.

8

3. If $f(x,y,z) = (x \oplus y')*z'$, use the diagram at the right to compute:

$f(a,d,c) =$ _____

$f(c,f,1) =$ _____



5

4. Prove the statement: "Every binary tree has the property that the number of nodes in that tree is exactly one more than the number of edges in that tree". (Hint: Write down the statement you will use for P(n).)

8

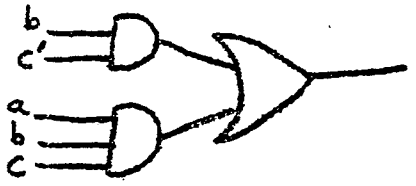
5. Write down one of the Absorption Laws (L4) for a Boolean Algebra, and then rigorously prove that law. (Identify each implication that you use).

7

6. Draw the Karnaugh map for the 5-variable function:
$\sum$ 0,1,2,8,10,15,16,17,18,20,21,22,24,25,26,28,29,30
Draw in the cubes.
Give the minimal representation of this function.

10

7a Draw a circuit which will cheaply and efficiently perform the same function as the circuit illustrated below: (Use the Quine McClusky algorithm)



b) Put the above formula into principle conjunctive normal form.

8. Using the diagram, answer the following:

find all the upper bounds of {d,e,f}:_____

GLB{a,b,c} = _____

the maximal element(s):_____

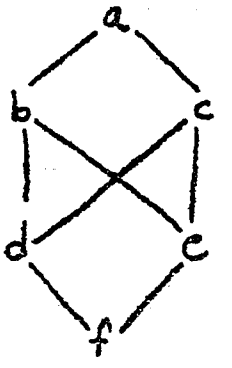
which elements have complements?_____

is this structure: a partial ordering?_____

a tree?_____

a Boolean Algebra?_____

a lattice?_____



9. Prove that if a relation R in X is antisymmetric, then $\tilde{R}$ is antisymmetric.

9

10. Show that the conclusion $(\exists x)M(x)$ can be deduced from the hypotheses $H_1: (\forall y)(H(y) \rightarrow M(y))$ and $H_2: (\exists z)H(z)$
<u>Identify each implication that you use.</u>

10

11. Write down the truth table for the NAND($\uparrow$) connective:

Support your answers to the following questions about NAND:
Is NAND commutative?

10

Does NAND distribute over AND(A)?

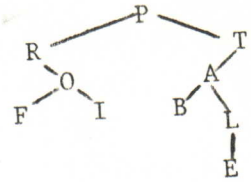
II

1 Determine the order in which the nodes of the tree at the right will be visited if the tree is traversed in preorder: _____

in postorder: _____

in inorder: _____

(10)



2 Write ((A+B)-C) x ((D-E) x F) in Polish notation: _____

Write the following prefix formula in completely parenthesized form:
++Ax-BCxD-EFG

Give the tree structure of this last formula:

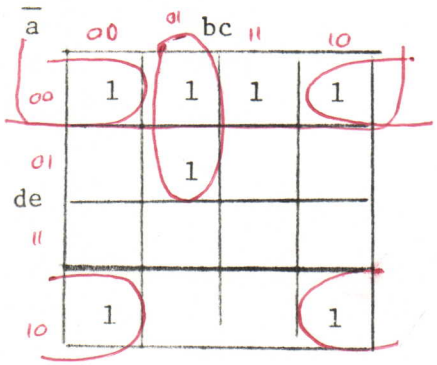
(10)

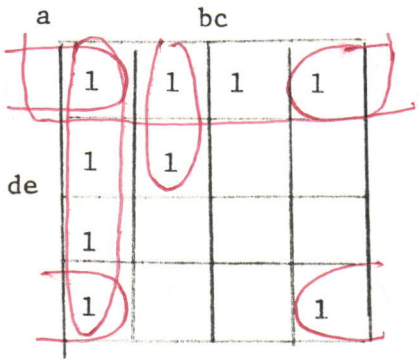
✓ 3 If f(x,y,z) = (x*y') ⊕ z, use the diagram at the right to compute: f(a,0,f)= (a*0')⊕f = (a*1)⊕f = a⊕f = 1

f(d,a,c)= (d*a')⊕c = (d*f)⊕c = b⊕c = f

(5)

4 For the function illustrated in the Karnaugh map below, circle the cubes and draw the corresponding Quine-McClusky chart. Determine the prime implicants and then draw the reduced chart. Use this to determine a minimal representation of the function.





(10)

$d'e' + b'cd' + ab'c' + c'e'$

5 If f and g are functions whose domain is the set of real numbers, and are given by the formulas at the right, find expressions for the following functions or explain why they do not exist.

$f(x) = 3x^2$

$g(x) = 2x - 4$

$f \circ g\ (x) =$ _____

$g \circ f\ (x) =$ _____

$g^{-1}(x) =$ _____

$f^{-1}(x) =$ _____

10

✓ 6 Put the following formula into principle conjunctive normal form:
$\neg((P \uparrow Q) \rightarrow (\neg P \wedge R))$

10

b) use a) to write the formula in Π notation: _____

c) Use b) to write the formula in Σ notation: _____

d) use c) to write the formula in principle disjunctive normal form:

✓ 7 Does $\rightarrow$ distribute over $\vee$? (Prove or disprove without using truth tables - identify the equivalences that you use)

10

8 Show that the conclusion $C: P \rightarrow S$ can be deduced from the hypotheses H_1: $\neg P \vee Q$, H_2: $\neg Q \vee R$, and H_3: $R \rightarrow S$. Identify each implication that you use.

10

1) P assumed
2) $\neg P \vee Q$ P
3) Q
4) $\neg Q \vee R$ P
5) R
6) $R \rightarrow S$ H_3 P
7) S I_{11} (5,6)
8) $P \rightarrow S$ CP (1,7)

9 Illustrate a one-to-one correspondence between the set of <u>all</u> integers and the set of positive integers.

5

10a) The operations of * and ⊕ in any Boolean algebra are duals of each other, since they satisfy DeMorgan's laws. Write the GeneralizedDeMorgan's equivalen of the formula: (A*B') ⊕ C'

_____

b) For the Boolean algebra consisting of $\langle \rho(E), \subseteq \rangle$, give the corresponding formula using set theory notation:

10

$\{\langle a,b\rangle \mid a \subseteq b,\ a \in \rho(E),\ b \in \rho(E)\}$

11 Given the relation R = {<x,y> | x+y is even } over the set {0,1,2,3,4} .

Draw the graph of R:

Find M_R:

Is R reflexive?

Is R symmetric?

Is R transitive?

Is R an equivalence relation?

(Give reasons for your answers.)

15

12 Prove that if $A \subseteq B$, then $A \cap (B \cup C) = A$.

(10)

13 Prove that the converse of a partial order relation over S is also always a partial order relation over S.

(10)